

EXERCISE 01 · S Q L F OUNDATIONS SQL

Fundamentals: SELECT & Filtering

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Question 01: Retrieve all columns from the employees table. EXPECTED COLUMNS: id first_name last_name department salary hire_date city

```
25 SELECT * FROM sorting.default.employees;
26
27 --Question 02: Find all unique departments
28
29 SELECT DISTINCT department FROM sorting.default.employees;
30 --Question 03: Retrive first and last names ordered by salary desc
31
32 SELECT first_name, last_name FROM sorting.default.employees ORDER BY salary DESC;
```

Add parameter

Table							
	employee_id	first_name	last_name	department	salary	hire_date	city
1	1	Alice	Green	IT	70000	2020-01-10	Johannesbu...
2	2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	3	Cathy	Zulu	FINANCE	65000	2018-07-18	Durban
4	4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	5	Eva	Naidoo	IT	72000	2017-09-30	Johannesbu...

Question 02: Find all unique departments. EXPECTED COLUMNS: department

```
28
29 SELECT DISTINCT department FROM sorting.default.employees;
30 --Question 03: Retrive first and last names ordered by salary desc
31
32 SELECT first_name, last_name FROM sorting.default.employees ORDER BY salary DESC;
```

Add parameter

Table							
	department						
1	IT						
2	HR						
3	FINANCE						
4	Marketing						

Question 3: Retrieve first and last names ordered by salary descending.
EXPECTED COLUMNS: first_name last_name

--Question 03: Retrieve first and last names ordered by salary desc

```
SELECT first_name, last_name FROM sorting.default.employees ORDER BY salary DESC;
```

--Question 04: Retrieve the top 3 highest-paid employees

```
SELECT employee_id, first_name, last_name, salary FROM sorting.default.employees ORDER BY
```

parameter

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A _C first_name	A _C last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

Question 04: Retrieve the top 3 highest-paid employees. EXPECTED COLUMNS: id first_name last_name salary

--Question 04: Retrieve the top 3 highest-paid employees

```
SELECT employee_id, first_name, last_name, salary FROM sorting.default.employees ORDER BY salary DESC LIMIT 3;
```

--Question 05: Find employees in the IT department

parameter

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employee_id	A _C first_name	A _C last_name	salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

```
38 --Question 05: Find employees in the IT department
39
40 SELECT employee_id, first_name, last_name, department FROM sorting.default.employees WHERE department = 'IT';
41
42 --Question 06: Find employees in finance with salary > 60000
```

	^A ₂ ^B ₃ employee_id	^A ₁ ^B ₂ first_name	^A ₁ ^B ₂ last_name	^A ₁ ^B ₂ department
1	1	Alice	Green	IT
2	5	Eva	Naidoo	IT

```

42 --Question 06: Find employees in finance with salary > 60000
43
44 SELECT employee_id , first_name, last_name, department, salary FROM sorting.default.employees WHERE department =
45 'FINANCE' AND salary > 60000;
46
47 --Question 07: Find employees in HR or Marketing

```

	$\mathbb{A}_3^2$ employee_id	$\mathbb{A}_C^B$ first_name	$\mathbb{A}_C^B$ last_name	$\mathbb{A}_C^B$ department	$\mathbb{A}_3^2$ salary
1	3	Cathy	Zulu	FINANCE	65000

```
SELECT employee_id, first_name, last_name, department FROM sorting.default.employees WHERE department = 'HR' OR department = 'Marketing';
```

```
SELECT employee_id, first_name, last_name, department FROM sorting.default.employees WHERE department <> 'IT';
```

parameter

employee_id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

Question 8: Find employees not in IT. EXPECTED COLUMNS: id first_name last_name department

```
0
1  --Question 08: Find employees not in IT
2
3  SELECT employee_id, first_name, last_name, department FROM sorting.default.employees WHERE department <> 'IT';
4
5  --Question 09: Find employees in IT, HR, or Finance using IN
6
7  SELECT employee_id, first_name, last_name, department FROM sorting.default.employees WHERE department IN ('IT', 'HR',
  'FINANCE');
```

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	123 employee_id	A ^B first_name	A ^B last_name	A ^B department					
	2	Brian	Lee	HR					
	3	Cathy	Zulu	FINANCE					
	4	David	Mokoena	Marketing					

Question 9: Find employees in IT, HR, or Finance using IN. EXPECTED COLUMNS: id first_name last_name department

```
54
55  --Question 09: Find employees in IT, HR, or Finance using IN
56
57  SELECT employee_id, first_name, last_name, department FROM sorting.default.employees WHERE department IN ('IT', 'HR',
```

Add parameter

Table ▾ +					🔍 🏠 📄 📁 ⌵ ✕				
	123 employee_id	A ^B first_name	A ^B last_name	A ^B department					
1	1	Alice	Green	IT					
2	2	Brian	Lee	HR					
3	3	Cathy	Zulu	FINANCE					
4	5	Eva	Naidoo	IT					

Question 10: Find employees in IT with salary > 65000 and city Johannesburg.
EXPECTED COLUMNS: id first_name last_name department salary city

```
58
59 --Question 10: Find employees in IT with salary > 65000 and city Johannesburg
60
61 SELECT employee_id, first_name, last_name, department, salary, city FROM sorting.default.employees WHERE department =
    'IT' AND salary > 65000 AND city = 'Johannesburg';
62
63 --Question 11
```

Add parameter

Table								
	employee_id	first_name	last_name	department	salary	city		
1	1	Alice	Green	IT	70000	Johannesbu...		
2	5	Eva	Naidoo	IT	72000	Johannesbu...		